Propane Gas Sensor

SMD1011 Product Datasheet

1. Product Introduction

The SMD1011 propane sensor is a semiconductor gas sensor developed based on MEMS technology. It can be used to detect the content of propane gas, and it has anti-interference to common organic gases such as alcohol.

The propane gas sensing material and sensor electrodes are developed and made by IDM. When propane gas contacts the sensitive material, the conductivity of the material will change. A specific circuit can be used to convert the change of conductivity into an output signal corresponding to the gas concentration.

2. Sensor Characteristics

- Very small size
- Low power consumption
- High sensitivity
- Fast response and recovery time
- Simple test circuit
- Good stability
- Long life time

3. Main Application

- Propane gas leak monitoring devices and firefighting/safety monitoring systems for homes, factories, and businesses
- Flammable gas leak alarm, line gas leak detector, smart gas meter, gas water heater, gas integrated stove, kitchen hood, etc

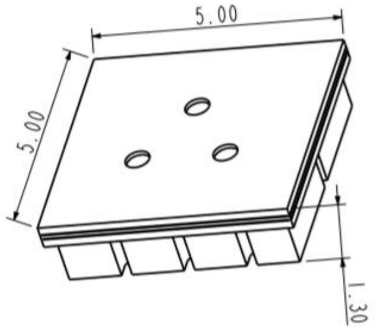
4. Product Description

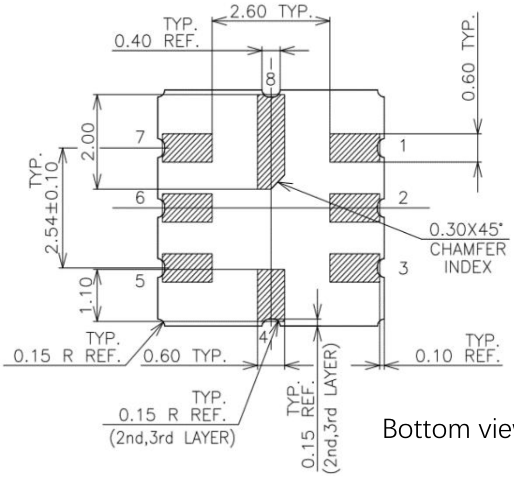
4.1 Technical parameters

Table 1

Product Number			SMD1011
Standard Package			Ceramic Package
Target Gas			Propane
Measurement Range			0 ~ 1000 ppm
Resolution			200 ppm
Standard Circuit Conditions	Sensor Voltage	V_C	≤ 5 V
	Heating Voltage	V_H	1.8 ± 0.05 V DC
	Load Resistance	R	Adjustable (subject to shipping sheet)
Gas Sensor Characteristics Under Standard Test Conditions	Heater Resistance	R_H	43 ±5 Ω (at room temperature)
	Heater Power Consumption	P_H	≤ 30 mW
	Sensor Resistance	R_S	5 ~ 300 KΩ (tested in air and standard condition)
	Sensitivity	S	R_0 (in air)/Rs (in 3000ppm propane) ≥3
	Response Ratio	α	≤ 0.6 (R 3000 ppm / R 1000ppm propane)
Standard Test Conditions	Temperature Humidity		20±2 ℃ 55±5%RH
	Preheat Time		≥ 3 min
Response Time (T_{90})			<15 s
Recovery Time (T_{10})			<60 s
Life Time			3 years

4.2 Sensor structure and pin Definition Diagram





Bottom view

Table 2

Pin Number	Name	Definition
1	VH+	Heating voltage supply
2	VCC	Sensor voltage supply
3	NG	/
4	NG	/
5	NG	/
6	NG	/
7	HOT	Heater GND
8	VOUT	Sensor output voltage

4.3 Basic Circuit

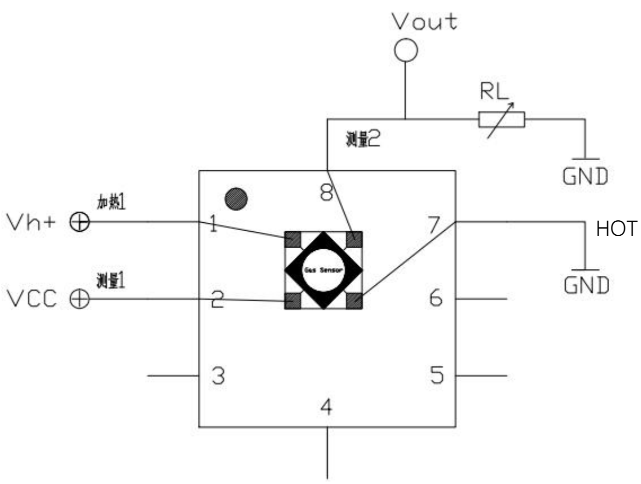


Figure shows the basic test circuit for SMD1001 sensor. The sensor needs to apply 2 voltages: heater voltage

Top view

(Vh) and sensor voltage (Vcc). Vh+ is a specific voltage for the sensor heating element. Vcc is the sensor voltage. Vout is output reading voltage which need connect the load resistor RL in series. These two voltage need set in right value according to datasheet, otherwise measurement performance will reduce and even damage the sensor.

5. Sensor Characterization

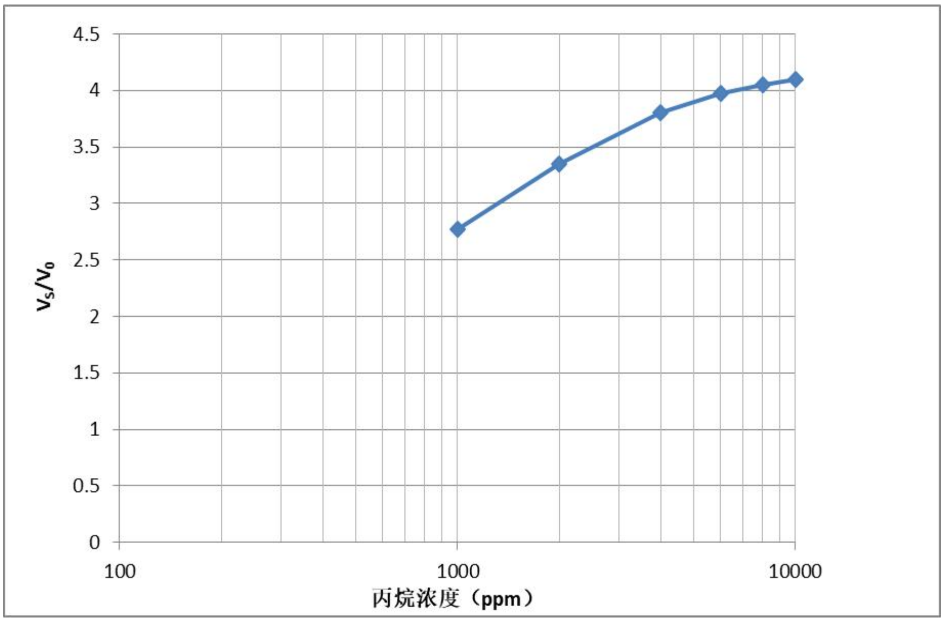


Figure 1 Logarithmic characteristic curve of sensor sensitivity

In above figure, the test was completed under standard test conditions, and the load resistance used was 10kΩ. V0 represents the test voltage of the which tested in clean air; and Vs represents the test voltage of the sensor in target gas in different concentrations.

Note: in following section, if the unit is resistance, it is calculated by this formula:

$$R = \frac{R_L(V_{CC} - V_{OUT})}{V_{OUT}}$$

Each item definition refer table1

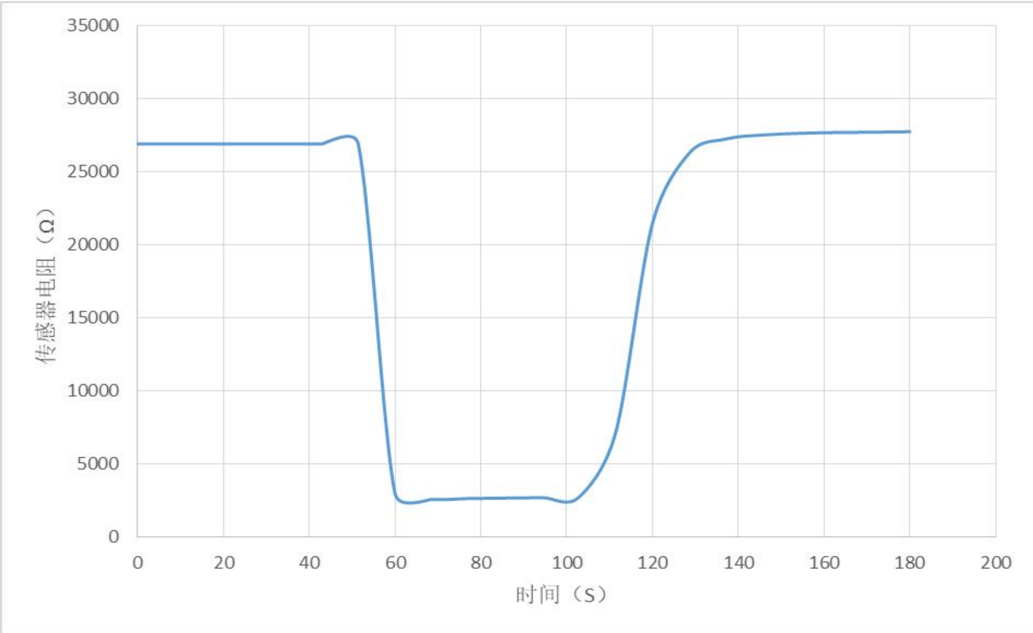


Figure 2 Sensor response-recovery characteristic curve

Above figure shows the resistance change characteristic curve of the sensor when the test gas is 2000ppm propane. All tests are completed under standard test conditions.

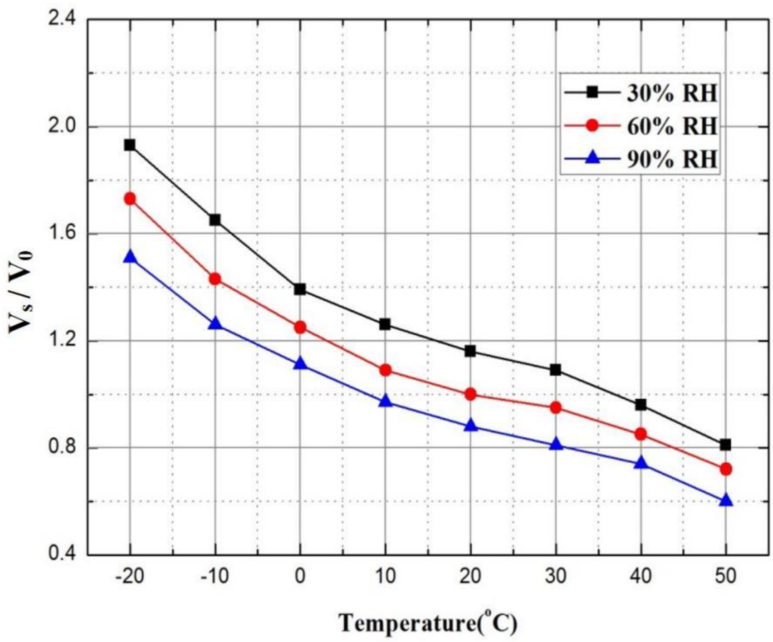


Figure 3 Temperature and humidity characteristic curve of the sensor

Above figure shows the temperature and humidity effect curves, Vs represents the response voltage in 2000 ppm propane gas under different temperature and humidity conditions, and V_0 represents the response voltage value in clean air under correspond temperature and humidity conditions.

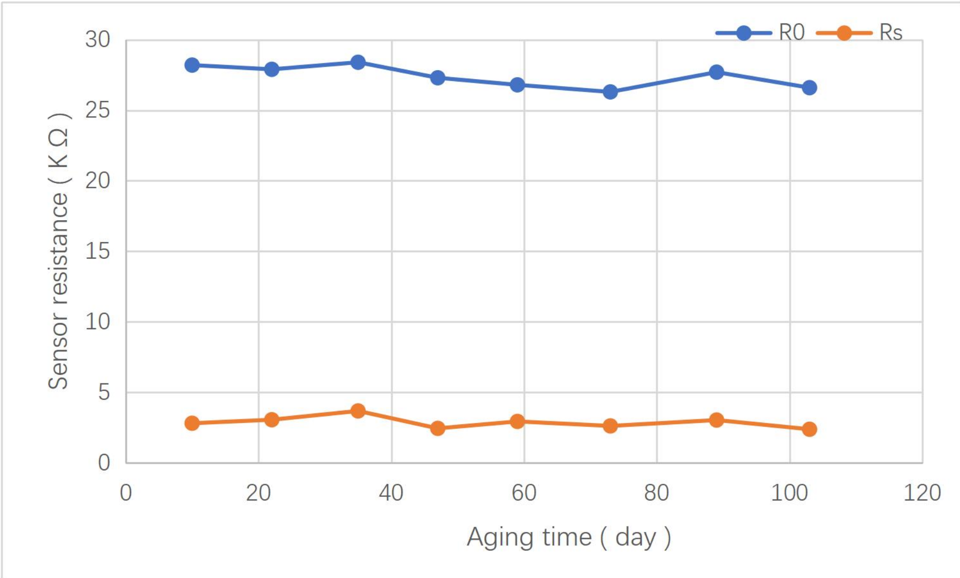


Figure 4 Long-term stability test of SMD1011 sensor

All tests in the figure are completed under standard test conditions. R_0 represents the resistance of the sensor in clean air, Rs represents the resistance value of the sensor in 2000ppm propane gas, the horizontal axis is the test time of continuous power-on, and the vertical axis is the actual resistance value of the sensor.

6. Product Shipping Packaging:

Generally shipped in carrier tape packaging. Customer can request other package when place the purchase.

Precautions:

Situations to avoid

1.1 Exposure to volatile silicon compound vapor

The sensor should avoid being exposed to silicone adhesives, hair spray, silicone rubber, putty or other places where volatile silicone compounds exist. If the surface of the sensor is adsorbed with silicone compound vapor, the sensitive material of the sensor will be wrapped by silicon dioxide which formed by the decomposition of the silicone compound, which will degrade the sensitivity of the sensor and cannot be restored..

1.2 Highly corrosive environment

The sensor is exposed to high concentration of corrosive gas (such as H2S, SOX , Cl 2 , HCl, etc.), it will not only cause corrosion or damage to the heating material and sensor leads, but also cause irreversible deterioration of the performance of sensing materials.

1.3 Pollution by alkali, alkali metal salts and halogens

The performance of the sensor may also deteriorate when it is contaminated by alkali metals, especially salt water spray, or exposed to halogens such as Freon.

1.4 Directly tough with water

Splashing or immersing the sensor in water will cause the sensor's sensitivity to decrease.

1.5 Freeze

If sensor surface appear with ice, it will cause sensing material crack and lost sensitive property.

1.6 Applied voltage is too high

If the voltage applied to the sensor or heater is higher than the specified value, even if the sensor is not physically damaged or destroyed, it may cause damage to the leads and/or heater and cause the sensor's sensitivity characteristics to degrade.

1.7 Voltage pins connect mismatch

If the wrong voltage is applied to the sensor or heating and signal pins, it may cause damage to the leads and/or heater and cause the sensor's sensitivity to deteriorate.

2. Situations need attention

2.1 Condensation

Under indoor use conditions, slight condensation will have a slight effect on sensor performance. However, if water condenses on the surface of the sensitive layer and remains for a long time, the sensor characteristics will deteriorate.

2.2 In high concentration gas

Regardless of whether the sensor is powered on or not, long-term exposure to high-concentration gas will affect the sensor's characteristics. For example, spraying lighter gas directly at the sensor will cause great damage to the sensor.

2.3 Long-term storage

If the sensor is stored for a long time without power, its resistance will have a reversible drift, which is related to the storage environment. The sensor should be stored in a sealed bag that does not contain volatile silicon compounds. After long-term storage, the sensor needs to be powered for a longer time before use to stabilize it. The storage time and corresponding aging time are recommended as following:

Storage time	Recommended aging time
Less than 1 month	Not less than 6 hours
1-6 months	Not less than 12 hours
Moe than 6 months	No less than 24 hours

2.4 Long-term exposure to extreme environments

Regardless of whether the sensor is powered or not, prolonged exposure to extreme conditions such as high humidity, high temperature or high pollution will seriously affect sensor performance.

2.5 Vibration

Frequent, excessive vibration can cause the sensor's internal leads to resonate and break. This type of vibration can be generated during transportation and by using pneumatic screwdrivers/ultrasonic welders on the assembly line.

2.6 Excessive impact

If the sensor is subjected to a strong impact or falls, its lead wires may break.

2.7 Conditions of assembling:

2.7.1 Manual welding is the most ideal welding method for sensors. The recommended welding conditions are as following:

Flux: Rosin flux with minimal chlorine content

Constant temperature soldering iron

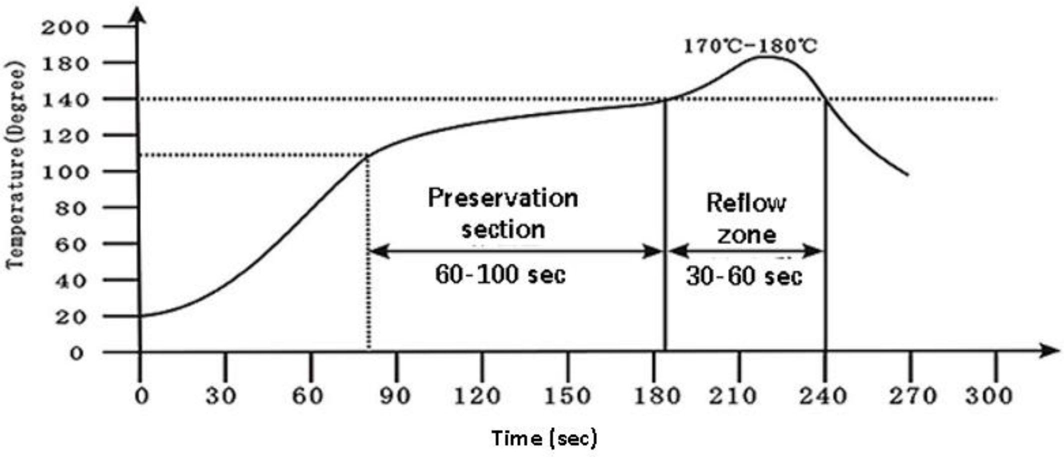
Temperature: 250℃

Time: no more than 3 seconds

2.7.2 The following conditions are recommended when using reflow soldering :

Solder paste: low temperature lead-free solder paste (Sn42Bi58)

The furnace curve is as following:



2.8 Anti-static

Anti-static bag packaging

Violation of the above usage conditions will degrade the sensor characteristics.

Suzhou Huiwen Nanotechnology Co., Ltd.

http://www.huiwen-sensor.com/

Room 302, Building 17, No. 99 , Jinji Lake Avenue,
Suzhou Industrial Park, Jiangsu Province, China

Tel/Whatsapp:008618868807612

E-Mail :overseassales@idmsensor.com